A NOTE ON PROBLEM #650

1. Introduction

For each positive integer m, let $f(m)$ be the largest integer with the following property: whenever

$$A = \{a_1 < \cdots < a_m\}$$

is a set of m positive integers and $x \in \mathbb{R}$, the open interval

$$I = (x, x + 2a_m)$$

contains distinct integers $b_1, \ldots, b_r$ and distinct elements $c_1, \ldots, c_r \in A$ such that

$$c_i \mid b_i \qquad (1 \le i \le r),$$

with

$$r \ge f(m).$$

In this note, we determine $f(m)$ for all $m \ge 4$, proving that

$$f(m) = \lceil 2\sqrt{m} \rceil \qquad (m \ge 4).$$

2. Upper bound

Theorem 2.1. *For all integers $s, t \ge 1$, one has*

$$f(st) \le s + t.$$

Consequently, for every $m \ge 4$,

$$f(m) \le \lceil 2\sqrt{m} \rceil.$$

Proof. We first prove that

$$f(st) \le s + t \qquad (s, t \ge 1).$$

If $s = 1$ or $t = 1$, then trivially

$$f(st) \le st \le s + t,$$

so we may assume $s, t \ge 2$.

Define

$$E := \prod_{p \le st} p^{1 + \lfloor \log_p(s-1) \rfloor}, \qquad D := E \cdot \operatorname{lcm}(1, 2, \ldots, st).$$

Then for every prime $p \le st$ and every integer δ with

$$1 \le |\delta| \le s - 1,$$

we have

$$v_p(D) > v_p(\delta),$$

where $v_p(\cdot)$ denotes the p-adic valuation. Indeed,

$$v_p(\delta) \le \lfloor \log_p(s-1) \rfloor,$$

whereas the exponent of p in E is

$$1 + \lfloor \log_p(s-1) \rfloor.$$

Date: March 7, 2026.

Next let

$$\mathcal{P} := \Big\{ p > st : \ p \text{ is prime and } p \mid (qD + \delta) \text{ for some } 1 \le q \le t-1, \ 1 \le |\delta| \le s-1 \Big\}.$$

This is a finite set of primes.

For each $p \in \mathcal{P}$, consider the residue classes

$$i - jD \pmod p \qquad (1 \le i \le s, \ 0 \le j \le t-1).$$

There are at most st such classes, whereas $p > st$, so they do not exhaust all residue classes modulo p. Hence we may choose a residue class

$$r_p \pmod p$$

such that

$$r_p \not\equiv i - jD \pmod p \qquad (1 \le i \le s, \ 0 \le j \le t-1).$$

By the Chinese remainder theorem, there exists an integer M such that

$$M \equiv r_p \pmod p \qquad (\forall p \in \mathcal{P}),$$

and also

$$M > 2(t-1)D + 4s.$$

For $1 \le i \le s$ and $0 \le j \le t-1$, define

$$\alpha_{i,j} := M + jD - i.$$

These integers are pairwise distinct. Indeed, if

$$M + jD - i = M + \ell D - k,$$

then

$$(j - \ell)D = i - k.$$

Since

$$|i - k| < s \le st \le D,$$

we must have $j = \ell$, and then $i = k$.

Now set

$$A := \{\alpha_{i,j} : 1 \le i \le s, \ 0 \le j \le t-1\}, \qquad N := M + (t-1)D - 1.$$

Then $|A| = st$, and the largest element of A is exactly N, since

$$\alpha_{1,t-1} = M + (t-1)D - 1 = N.$$

We claim that for any two pairs (i, j) and (k, ℓ),

$$\gcd(\alpha_{i,j}, \alpha_{k,\ell}) \mid (k - i).$$

Let

$$d := \gcd(\alpha_{i,j}, \alpha_{k,\ell}).$$

Since

$$\alpha_{i,j} - \alpha_{k,\ell} = (j - \ell)D + (k - i),$$

we have

$$d \mid (j - \ell)D + (k - i).$$

There is nothing to prove if $i = k$, and if $j = \ell$ then immediately

$$d \mid (k - i).$$

So assume $i \ne k$ and $j \ne \ell$. Put

$$q := |j - \ell| \in \{1, \dots, t-1\},$$

and define

$$\delta = \begin{cases} k-i, & j > \ell, \\ i-k, & j < \ell. \end{cases}$$

Then

$$d \mid qD + \delta, \qquad 1 \le |\delta| = |k-i| \le s-1.$$

We first show that every prime divisor of d is at most st. Suppose, for contradiction, that some prime $p > st$ divides d. Since

$$p \mid d \mid qD + \delta,$$

we have $p \in \mathcal{P}$ by definition of $\mathcal{P}$. On the other hand,

$$p \mid d \mid \alpha_{i,j} = M + jD - i,$$

so

$$M \equiv i - jD \pmod{p},$$

contradicting the choice of $M \equiv r_p \pmod{p}$.

Thus every prime divisor of d is at most st. Now let $p \le st$ be any prime. Since

$$v_p(qD) \ge v_p(D) > v_p(\delta),$$

it follows that

$$v_p(qD + \delta) = v_p(\delta).$$

Because $d \mid qD + \delta$, we obtain

$$v_p(d) \le v_p(qD + \delta) = v_p(\delta) = v_p(k-i).$$

Since all prime divisors of d are at most st, we conclude that

$$d \mid (k-i).$$

This proves the claim.

Consequently, the system of congruences

$$x \equiv -i \pmod{\alpha_{i,j}} \qquad (1 \le i \le s,\ 0 \le j \le t-1)$$

is pairwise compatible: for any two congruences in the system, the difference of the residues is

$$(-i) - (-k) = k - i,$$

and this is divisible by $\gcd(\alpha_{i,j}, \alpha_{k,\ell})$. Hence, by the generalized Chinese remainder theorem, the system has a solution, say x_0.

Choose an integer T such that

$$2\big((t-1)D + s - 1\big) < T \le M - 2s.$$

Such a choice is possible because

$$M > 2(t-1)D + 4s.$$

Consider the open interval

$$I := (x_0 - T,\ x_0 - T + 2N).$$

Fix (i,j) with $1 \le i \le s$ and $0 \le j \le t-1$. Since

$$x_0 \equiv -i \pmod{\alpha_{i,j}},$$

we have

$$\alpha_{i,j} \mid (x_0 + i).$$

Also,

$$x_0 + M + jD = (x_0 + i) + \alpha_{i,j},$$

so $x_0 + M + jD$ is another multiple of $\alpha_{i,j}$.

We now show that these are exactly the two multiples of $\alpha_{i,j}$ contained in I.

First, both of them lie in I. Indeed, the offset of $x_0 + i$ from the left endpoint is

$$T + i,$$

which is positive and satisfies

$$T + i \leq T + s \leq M - s < 2N.$$

Hence

$$x_0 + i \in I.$$

Similarly, the offset of $x_0 + M + jD$ from the left endpoint is

$$T + M + jD,$$

which is positive. Moreover,

$$T + M + jD \leq T + M + (t-1)D \leq (M - 2s) + M + (t-1)D < 2N,$$

because

$$2N = 2M + 2(t-1)D - 2$$

and $s, t \geq 2$.

Next, the preceding multiple of $\alpha_{i,j}$ is

$$(x_0 + i) - \alpha_{i,j},$$

whose offset from the left endpoint is

$$T + i - \alpha_{i,j}.$$

But

$$\alpha_{i,j} = M + jD - i \geq M - s,$$

and

$$T + i \leq T + s \leq M - s \leq \alpha_{i,j},$$

so

$$T + i - \alpha_{i,j} \leq 0.$$

Hence this preceding multiple does not belong to I.

Finally, the next multiple after $x_0 + M + jD$ is

$$(x_0 + i) + 2\alpha_{i,j},$$

whose offset from the left endpoint is

$$T + i + 2\alpha_{i,j} \geq T + 1 + 2(M - s).$$

Using

$$T > 2\big((t-1)D + s - 1\big) = 2(t-1)D + 2s - 2,$$

we obtain

$$T + i + 2\alpha_{i,j} > 2(t-1)D + 2s - 2 + 1 + 2M - 2s = 2M + 2(t-1)D - 1 > 2N,$$

because

$$2N = 2M + 2(t-1)D - 2.$$

Therefore this next multiple also lies outside I.

Thus $\alpha_{i,j}$ has exactly two multiples in I, namely

$$x_0 + i \qquad \text{and} \qquad x_0 + M + jD.$$

It follows that every integer in I divisible by some element of A lies in

$$B := \{x_0 + i : 1 \leq i \leq s\} \cup \{x_0 + M + jD : 0 \leq j \leq t-1\}.$$

These two parts are disjoint, because

$$x_0 + i = x_0 + M + jD$$

would imply

$$i = M + jD,$$

which is impossible since $1 \le i \le s < M \le M + jD$. Hence

$$|B| = s + t.$$

Therefore, inside the interval I, there are at most $s + t$ integers divisible by some element of A. In particular, there cannot exist a matching from A into $I \cap \mathbb{Z}$ of size larger than $s + t$. Hence

$$f(st) \le s + t.$$

Now let $m \ge 1$, and put $n := \lceil 2\sqrt{m} \rceil \ge 4$. Choose

$$s := \left\lfloor \frac{n}{2} \right\rfloor, \qquad t := \left\lceil \frac{n}{2} \right\rceil.$$

Then

$$s + t = n = \lceil 2\sqrt{m} \rceil$$

and

$$st = \left\lfloor \frac{n^2}{4} \right\rfloor \ge m,$$

because $n \ge 2\sqrt{m}$.

Applying the construction above, we obtain a set A of size st whose largest element is N, together with an open interval of length $2N$, such that at most $s + t$ integers in that interval are divisible by some element of A.

Choose any subset $A' \subseteq A$ with

$$|A'| = m \qquad \text{and} \qquad N \in A'.$$

Then $\max A' = N$, and the same interval has length $2 \max A'$. Moreover, the set of integers in that interval divisible by some element of A' is contained in the corresponding set for A, so it still has cardinality at most $s + t$. Therefore

$$f(m) \le s + t = \lceil 2\sqrt{m} \rceil.$$

This completes the proof. □

3. Lower bound

Theorem 3.1. *Let $m \ge 4$, let*

$$A = \{a_1 < \cdots < a_m\}, \qquad N := a_m,$$

and let

$$I = (x, x + 2N).$$

Then there exist distinct

$$c_1, \ldots, c_L \in A \qquad \text{and} \qquad b_1, \ldots, b_L \in I \cap \mathbb{Z}$$

such that

$$c_i \mid b_i \qquad (1 \le i \le L),$$

where

$$L := \lceil 2\sqrt{m} \rceil.$$

Consequently,

$$f(m) \ge \lceil 2\sqrt{m} \rceil \qquad (m \ge 4).$$

Proof. We distinguish two cases.

Case 1: x is not an integer multiple of N.

Let

$$B := I \cap \mathbb{Z}, \qquad B_- := (x, x+N] \cap \mathbb{Z}, \qquad B_+ := (x+N, x+2N) \cap \mathbb{Z}.$$

Then

$$B = B_- \sqcup B_+.$$

Form the bipartite graph G with left vertex set A and right vertex set B, where $a \in A$ is joined to $b \in B$ if and only if $a \mid b$. For $S \subseteq A$, let $\Gamma(S)$ denote the neighbourhood of S in G, and put

$$\Gamma_-(S) := \Gamma(S) \cap B_-, \qquad \Gamma_+(S) := \Gamma(S) \cap B_+.$$

We claim that for every nonempty subset $S \subseteq A$,

$$|S| \le |\Gamma_-(S)|\,|\Gamma_+(S)|.$$

Fix such an S. For each $a \in S$, let u_a be the smallest multiple of a that is strictly greater than x, and let v_a be the largest multiple of a that is strictly less than $x + 2N$.

Since $a \le N$, we have

$$x < u_a \le x + a \le x + N,$$

so $u_a \in B_-$.

Also, if $a < N$, then

$$x + 2N - a < v_a < x + 2N,$$

hence

$$v_a > x + 2N - a > x + N,$$

so $v_a \in B_+$. If $a = N$, then $x \notin N\mathbb{Z}$ implies that $x + N$ is not a multiple of N; therefore the largest multiple of N below $x + 2N$ is strictly larger than $x + N$, and again $v_a \in B_+$. Thus every $a \in S$ has at least one multiple in B_- and at least one multiple in B_+.

Let c_a be the largest multiple of a lying in B_-. Then $c_a + a$ is the next multiple of a after c_a. Since $v_a \in B_+$ is also a multiple of a and satisfies $v_a > c_a$, we have

$$c_a + a \le v_a < x + 2N.$$

On the other hand, by maximality of c_a in B_-, the next multiple $c_a + a$ cannot belong to B_-, so

$$c_a + a > x + N.$$

Hence

$$c_a + a \in B_+.$$

Therefore

$$(c_a, c_a + a) \in \Gamma_-(S) \times \Gamma_+(S).$$

Define

$$\phi : S \to \Gamma_-(S) \times \Gamma_+(S), \qquad \phi(a) := (c_a, c_a + a).$$

This map is injective, because the difference of the two coordinates is exactly a:

$$(c_a + a) - c_a = a.$$

Hence

$$|S| \le |\Gamma_-(S)|\,|\Gamma_+(S)|.$$

Now let $S \subseteq A$ be nonempty, and write

$$s := |S|, \qquad r := |\Gamma(S)|.$$

Since $\Gamma_-(S)$ and $\Gamma_+(S)$ are disjoint subsets of $\Gamma(S)$, we have

$$|\Gamma_-(S)| + |\Gamma_+(S)| \le r.$$

By the elementary inequality

$$uv \le \left\lfloor \frac{(u+v)^2}{4} \right\rfloor \qquad (u, v \in \mathbb{Z}_{\ge 0}),$$

it follows that

$$s \le |\Gamma_-(S)|\,|\Gamma_+(S)| \le \left\lfloor \frac{r^2}{4} \right\rfloor.$$

Therefore

$$|\Gamma(S)| \ge \lceil 2\sqrt{|S|} \rceil \qquad \text{for every nonempty } S \subseteq A. \tag{3.1}$$

Since $m \ge 4$, we have $L = \lceil 2\sqrt{m} \rceil \le m$. Set

$$d := m - L \ge 0.$$

Adjoin d new vertices to the right-hand side of G, each adjacent to every vertex of A, and call the resulting bipartite graph G'.

We claim that Hall's condition holds in G'. For $t \ge 1$, define

$$h(t) := t - \lceil 2\sqrt{t} \rceil.$$

Because

$$0 < 2\sqrt{t+1} - 2\sqrt{t} < 1 \qquad (t \ge 1),$$

we have

$$\lceil 2\sqrt{t+1} \rceil - \lceil 2\sqrt{t} \rceil \le 1,$$

and hence

$$h(t+1) - h(t) = 1 - \left(\lceil 2\sqrt{t+1} \rceil - \lceil 2\sqrt{t} \rceil \right) \ge 0.$$

So h is nondecreasing on the positive integers.

Now let $S \subseteq A$ be nonempty. Using (3.1), we obtain

$$|S| - |\Gamma(S)| \le |S| - \lceil 2\sqrt{|S|} \rceil = h(|S|) \le h(m) = m - \lceil 2\sqrt{m} \rceil = d.$$

Hence

$$|\Gamma(S)| + d \ge |S|.$$

Since every dummy vertex is adjacent to every element of A, it follows that

$$|\Gamma_{G'}(S)| \ge |\Gamma(S)| + d \ge |S|.$$

The case $S = \varnothing$ is trivial. Thus Hall's theorem yields a matching in G' saturating all m vertices of A.

At most d edges of this matching can end at the d dummy vertices. After deleting those edges, we obtain a matching in the original graph G of size at least

$$m - d = L.$$

This proves the theorem in Case 1.

Case 2: x is an integer multiple of N.

Then N has exactly one multiple in I, namely

$$b_0 := x + N.$$

Set

$$A_0 := A \setminus \{N\}, \qquad B_- := (x, x+N) \cap \mathbb{Z}, \qquad B_+ := (x+N, x+2N) \cap \mathbb{Z},$$

and

$$B_0 := B_- \sqcup B_+ = (I \cap \mathbb{Z}) \setminus \{b_0\}.$$

Form the bipartite graph G_0 with left vertex set A_0 and right vertex set B_0, where $a \in A_0$ is joined to $b \in B_0$ if and only if $a \mid b$. For $S \subseteq A_0$, let $\Gamma_0(S)$ denote the neighbourhood of S in G_0.

We claim that for every nonempty subset $S \subseteq A_0$,

$$|\Gamma_0(S)| \geq \lceil 2\sqrt{|S|} \rceil.$$

Indeed, fix such an S. Since every $a \in S$ satisfies $a < N$, the smallest multiple of a strictly greater than x belongs to B_-, because

$$x < u_a \leq x + a < x + N.$$

Likewise, the largest multiple of a strictly less than $x + 2N$ belongs to B_+, because

$$x + 2N - a < v_a < x + 2N$$

and

$$x + 2N - a > x + N.$$

Now repeat verbatim the argument from Case 1: for each $a \in S$, if c_a denotes the largest multiple of a in B_-, then $c_a + a \in B_+$, and the map

$$a \mapsto (c_a, c_a + a)$$

is injective from S into $\Gamma_0(S) \cap B_- \ \times \ \Gamma_0(S) \cap B_+$. Therefore

$$|S| \leq |\Gamma_0(S) \cap B_-| \, |\Gamma_0(S) \cap B_+| \leq \left\lfloor \frac{|\Gamma_0(S)|^2}{4} \right\rfloor,$$

which implies

$$|\Gamma_0(S)| \geq \lceil 2\sqrt{|S|} \rceil.$$

If $m = 4$, then $|A_0| = 3$. For every subset $S \subseteq A_0$ we have

$$|\Gamma_0(S)| \geq \lceil 2\sqrt{|S|} \rceil \geq |S|,$$

because $|S| \leq 3$. Hence Hall's theorem gives a matching in G_0 saturating all three vertices of A_0. Together with the edge $N \sim b_0$, we obtain a matching of size

$$3 + 1 = 4 = L.$$

So it remains to consider $m \geq 5$. Set

$$L_0 := \lceil 2\sqrt{m-1} \rceil, \qquad d_0 := (m-1) - L_0.$$

Since $m - 1 \geq 4$, we have $L_0 \leq m - 1$, so $d_0 \geq 0$.

Adjoin d_0 new vertices to the right-hand side of G_0, each adjacent to every vertex of A_0, and call the resulting bipartite graph G_0'. Exactly as in Case 1, the bound

$$|\Gamma_0(S)| \geq \lceil 2\sqrt{|S|} \rceil \qquad (\varnothing \neq S \subseteq A_0)$$

implies Hall's condition in G_0'. Hence G_0' has a matching saturating all $m - 1$ vertices of A_0. Deleting the edges incident to the dummy vertices, we obtain a matching in G_0 of size at least

$$(m-1) - d_0 = L_0.$$

Now add the edge $N \sim b_0$. Because $b_0 \notin B_0$, this enlarges the matching by one. Hence we obtain a matching in the original graph of size at least

$$L_0 + 1 = \lceil 2\sqrt{m-1} \rceil + 1.$$

Finally,

$$0 < 2\sqrt{m} - 2\sqrt{m-1} < 1,$$

so

$$\lceil 2\sqrt{m} \rceil \leq \lceil 2\sqrt{m-1} \rceil + 1 = L_0 + 1.$$

Therefore the matching has size at least

$$L = \lceil 2\sqrt{m} \rceil.$$

This completes the proof in Case 2, and hence the theorem. □

Corollary 3.2. *For every integer $m \geq 4$,*

$$f(m) = \lceil 2\sqrt{m} \rceil.$$

Proof. The upper bound follows from Theorem 2.1, and the lower bound follows from Theorem 3.1. □